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COURSE: COAL

ASSIGNMENT NO 1

QUESTION 1

Describe the ENIAC computer in detail. Explain its architecture, technology used, and limitations. Compare ENIAC with modern computers in terms of speed, memory, and efficiency.

ENIAC Computer:

ENIAC (Electronic Numerical Integrator and Computer):

- Built in 1945 by John Presper Eckert and John Mauchly at the University of Pennsylvania.
- First general-purpose, electronic, digital computer.
- Designed for solving numerical problems for the U.S. Army (ballistic trajectories).

Architecture:

- Used decimal system (not binary).
- Consisted of 20 accumulators, each acting as a 10-digit register.
- Data flow was hardwired with plugboards and switches.
- They had separate units: accumulators, multipliers, dividers, function tables.

Technology Used:

- Based on vacuum tubes (about 18,000)
- Used relays, resistors, and capacitors
- Programmed manually (wiring and switches), no stored program

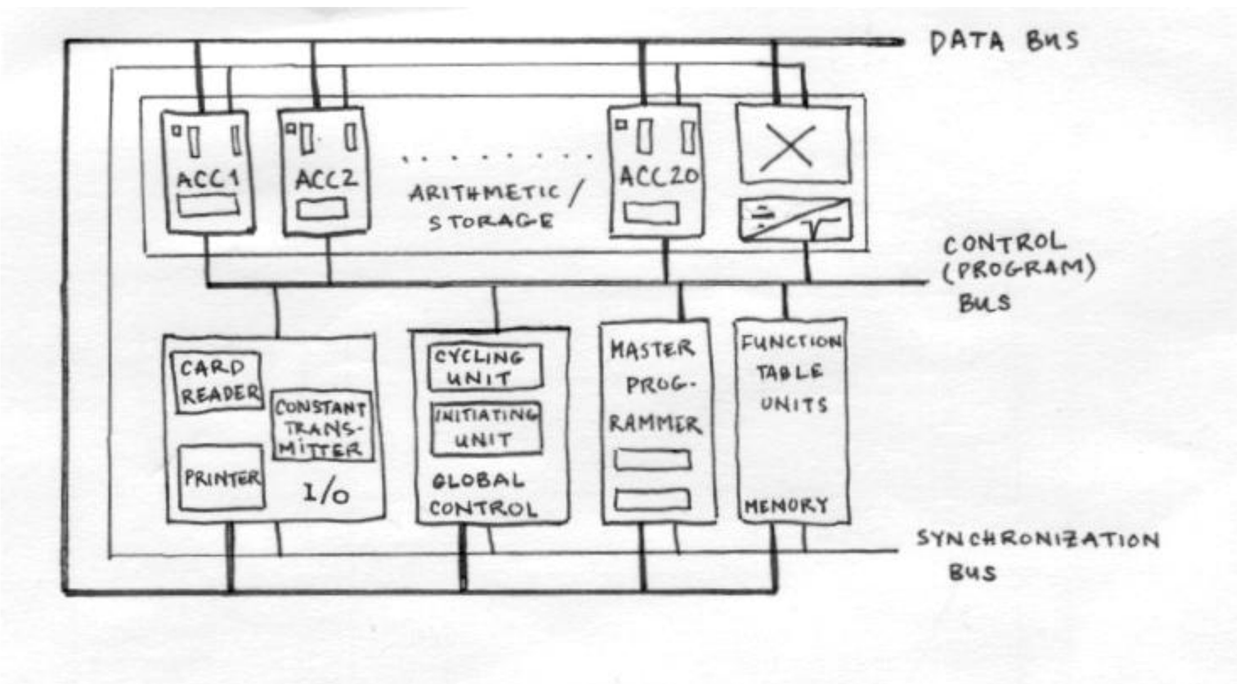
Limitations:

- Huge in size (30 tons, filled a large room)
- Consumed enormous power (~150 kW)
- Programming was slow and error-prone
- Limited memory capacity (only 20 numbers stored)

Comparison with Modern Computers:

Feature	ENIAC	Modern Computers
Speed	5,000 additions/sec	Billions/sec (GHz CPUs)
Memory	20 words (decimal)	Gigabytes/Terabytes
Efficiency	Consumed huge power	Energy efficient microchips
Programming	Manual wiring	Software (stored programs)

Diagram (Simplified ENIAC Unit Layout):



Reference:

Ceruzzi, P. (2003). *A History of Modern Computing*. MIT Press.

QUESTION 2

Explain the Von Neumann architecture. Draw and explain the structure of the IAS computer including all its registers. Discuss why the Von Neumann model is considered the foundation of modern computers.

<u>Von Neumann Architecture:</u>

Von Neumann Architecture (1945):

- Proposed by John von Neumann.
- Basis for most modern computers.
- Key principle: Stored Program Concept (program and data stored in same memory).

Components:

1. Memory (stores instructions + data).
2. Control Unit (CU) (fetches, decodes instructions).
3. Arithmetic Logic Unit (ALU) (performs operations).
4. Input/Output Devices.
5. Registers (fast small storage inside CPU).

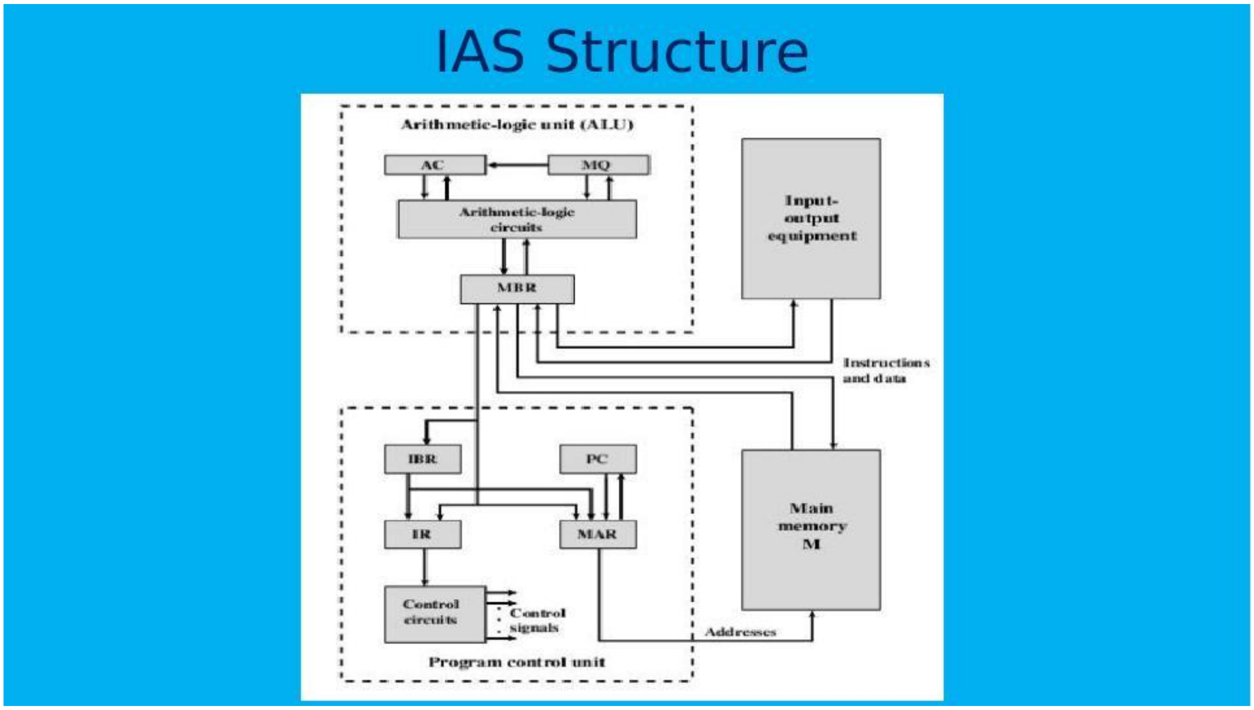
IAS Computer (Institute for Advanced Study, 1951):

- First implementation of Von Neumann architecture.

Registers in IAS Computer:

- Memory Buffer Register (MBR): Holds data fetched from memory.
- Memory Address Register (MAR): Holds address of memory location.
- Instruction Register (IR): Holds current instruction.
- Program Counter (PC): Holds address of next instruction.
- Accumulator (AC): For intermediate results.
- Multiplier Quotient (MQ): For multiplication/division.

Diagram (IAS Structure):



Why is it foundational?

- Clear separation of ALU, CU, Memory, and I/O.
- Flexible (programs easily modified).
- Still the blueprint of all modern computers.

Reference:

Von Neumann, J. (1945). *First Draft of a Report on the EDVAC*

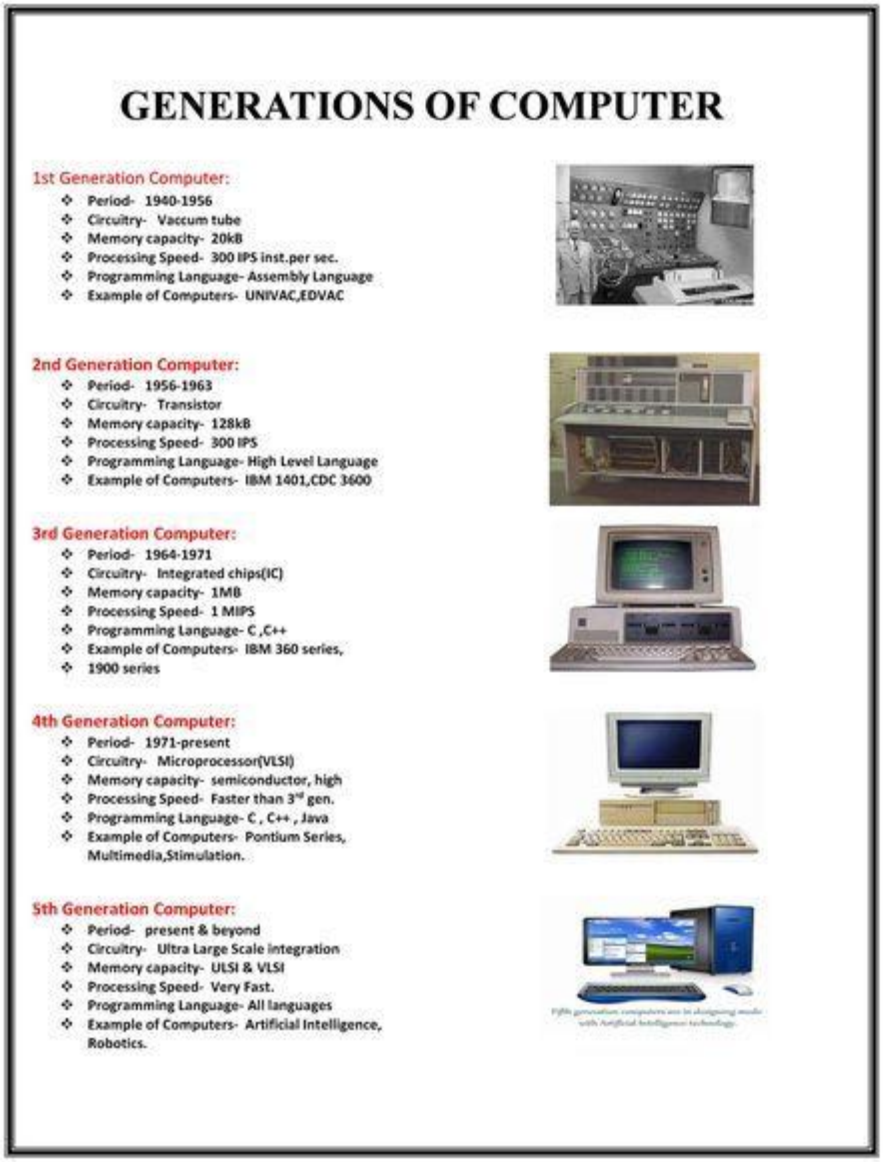
QUESTION 3

Discuss the different generations of computers (Vacuum tube, Transistor, SSI, MSI, LSI, VLSI, ULSI). For each generation, mention the main technology used, time period, advantages, and limitations.

<u>Generations of Computers:</u>

Generation	Time Period	Technology Used	Advantages	Limitations
1st	1940–1956	Vacuum Tubes	First electronic computers, fast compared to mechanical	Huge size, heat, unreliable
2nd	1956–1963	Transistors	Smaller, faster, less power	Still expensive, limited memory
3rd	1964–1971	SSI & MSI (Integrated Circuits)	More reliable, cheaper	Limited processing
4th	1971–1980s	LSI & VLSI (microprocessors)	Personal computers emerged	Dependent on silicon
5th	1980s–Present	ULSI, AI, parallel processing	Massive storage, AI, networking	Power dissipation, complexity
6th (Emerging)	Present–Future	Quantum, Nanotech	Super parallelism, AI-driven	Still experimental

Diagram:



Reference:

Mano, M. (2017). *Computer System Architecture*. Pearson.

QUESTION 4

Moore's Law has had a significant impact on computer technology. Explain Moore's Law and its consequences in detail. Do you think Moore's Law is still valid in today's context? Support your answer with examples.

<u>Moore's Law:</u>

Statement (1965 by Gordon Moore):

- Number of transistors on a chip doubles approximately every 18–24 months, leading to exponential growth in computing power.

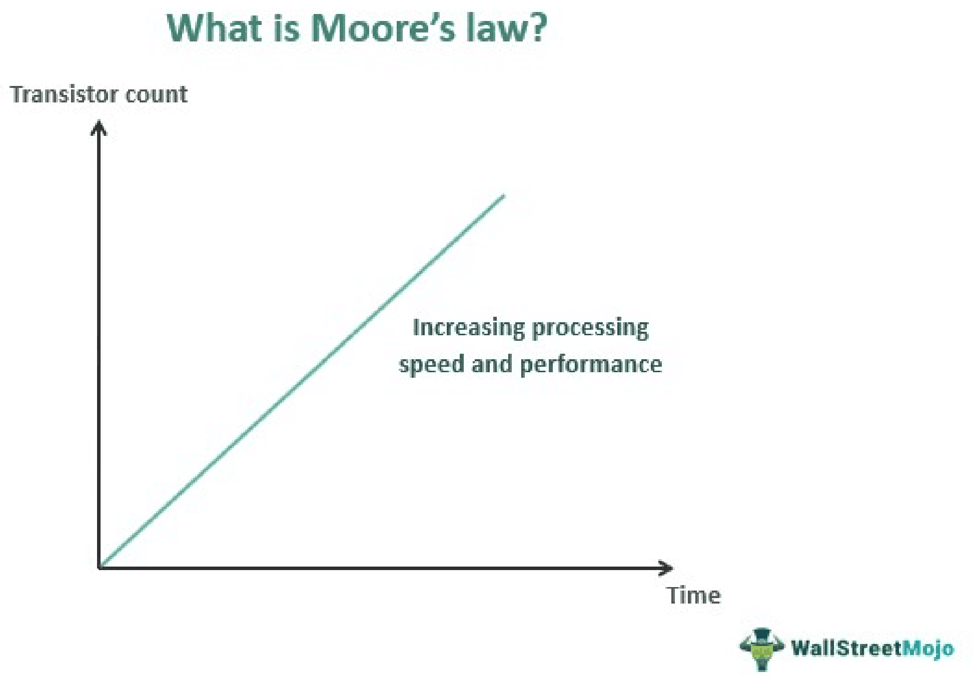
Consequences:

- Rapid growth in performance.
- Decreasing cost per transistor.
- Smaller, faster, more efficient devices.

Validity Today:

- Slowing down due to physical limits (nanometer-scale silicon).
- Examples:
    - Intel's 10m to 7mm nodes faced delays.
    - Chipmakers now use 3D stacking (chaplets), GPUs, and AI accelerators instead of just shrinking transistors.

Diagram:



Reference:

Schaller, R. R. (1997). *Moore's Law: Past, Present and Future*. IEEE Spectrum.

QUESTION 5

Compare and contrast UNIVAC I, IBM 701, and PDP-8 in terms of their design, purpose, and contribution to the computing industry.

UNIVAC I, IBM 701, and PDP-8

Feature	**UNIVAC I (1951)**	**IBM 701 (1952)**	**PDP-8 (1965)**
Design	First commercial computer, used mercury delay-line memory	IBM's first scientific computer, vacuum tubes	First successful minicomputer
Purpose	Business (census, payroll)	Scientific/military (defense simulations)	Affordable computing for labs, schools
Contribution	Start of commercial computing	Established IBM's dominance	Introduced small, cheap computers
Memory	1,000 words (12 chairs each)	2,048 words	4,096 words
Technology	Vacuum tubes	Vacuum tubes	Transistors

QUESTION 6

Write a short note on the evolution of Intel microprocessors from 4004 to Itanium 2. Explain the key features and performance improvements in each stage.

Intel Microprocessor Evolution (4004 → Itanium 2)

1) Intel 4004 (1971):

- 4-bit CPU, 2,300 transistors.
- First commercial microprocessor.

2) Intel 8008 (1972):

- 8-bit, 3,500 transistors.

3) Intel 8080 (1974):

- 8-bit, 6,000 transistors, used in Altair 8800.

4) Intel 8086 (1978):

- 16-bit, basis for x86 architecture.

5) Intel 80386 (1985):

- 32-bit introduced virtual memory.

6) Intel Pentium (1993):

- Superscalar, multiple pipelines.

7) Pentium Pro / Pentium II/III (1995–1999):

- Out-of-order execution, MMX instructions.

8) Pentium 4 (2000):

- Net Burst architecture, high clock speeds.

9) Itanium & Itanium 2 (2001–2002):

- 64-bit EPIC architecture, targeted servers.

Reference:

Intel Corporation. (2023). *Intel Processor History*.

QUESTION 7

Explain the concept of performance mismatch between processor and memory. Discuss techniques that have been used to overcome this issue (e.g., cache memory, pipelining).

<u>Processor–Memory Performance Mismatch</u>

Problem:

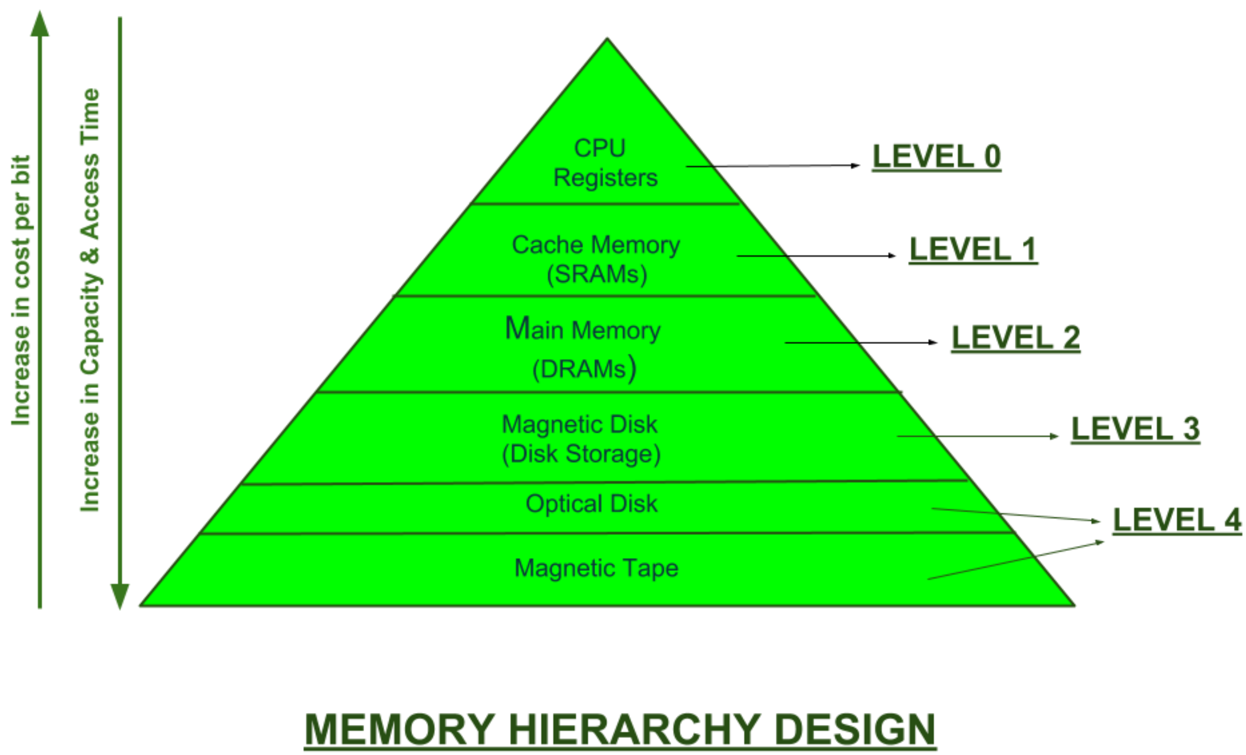
- CPU speed grows much faster than memory speed.
- Leads to "memory bottleneck".

Solutions:

1. Cache Memory (SRAM between CPU and main memory).

2. Pipelining (overlapping instruction execution).
3. Prefetching (loading instructions/data ahead).
4. Out-of-order execution.
5. Multi-level memory hierarchy (L1, L2, L3 caches).

Diagram (Memory Hierarchy)



QUESTION 8

Critically analyze the role of transistors and microelectronics in revolutionizing computer design. Why were these shifts necessary and how did they shape the industry?

Role of Transistors & Microelectronics

Why transistors were revolutionary:

- Smaller, faster, reliable, energy-efficient compared to vacuum tubes.
- Enabled mass production (integrated circuits).
- Paved way for miniaturization and microprocessors.

Impact:

- Birth of personal computers.
- Portable devices (laptops, smartphones).
- Rise of semiconductor industry (Intel, AMD, NVIDIA).
- Modern computing (AI, cloud, IoT) is only possible because of microelectronics.

Critical Analysis:

- Without transistors, computers would still be room sized.
- Microelectronics enabled exponential growth (Moore's Law).
- The shift was necessary to make computing affordable and universal.

Reference:

Riordan, M., & Hoddesdon, L. (1997). *Crystal Fire: The Birth of the Information Age*. Norton.